

Notes: Matějka & McKay (AER, 2015)

Rational Inattention to Discrete Choices: A New Foundation for the Multinomial Logit Model

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1 Big picture

- **Question.** Why do observed choices among discrete alternatives often look like multinomial logit probabilities, and can those probabilities be derived from optimal information acquisition rather than imposed as random utility errors?
- **Core idea.** A decision maker faces a discrete set of actions, does not initially know the true payoff vector, and can acquire information before choosing. Information is costly, so the decision maker rationally chooses an imperfect information structure.
- **Main result.** The optimal choice probabilities take a generalized multinomial logit form:

$$\mathbb{P}(i \mid \mathbf{v}) = \frac{P_i^0 e^{v_i/\lambda}}{\sum_{j=1}^N P_j^0 e^{v_j/\lambda}}, \quad (1)$$

where v_i is action i 's realized payoff, λ is the marginal cost of information, and P_i^0 is the unconditional probability of choosing action i before observing the state.

- **Connection to standard MNL.** If $P_i^0 = 1/N$ for all i , the formula becomes the standard multinomial logit. If prior beliefs make some actions more likely to be good, then the unconditional choice weights tilt probabilities toward those actions.
- **Interpretation of the scale parameter.** The logit scale parameter λ is not a black-box preference-noise parameter. It is the cost of processing one unit of information, measured using Shannon entropy.
- **Endogenous stochastic choice.** Stochastic choice does not come from exogenous idiosyncratic taste shocks. It comes from optimal limited attention: the decision maker sometimes chooses a non-best action because he does not acquire perfect information.
- **Prior beliefs matter.** Unlike the usual logit formula, choice probabilities depend on prior beliefs through the unconditional probabilities P_i^0 . Actions that look more promising ex ante receive more attention and are selected with higher unconditional probability.
- **IIA reinterpretation.** The model clarifies which form of independence of irrelevant alternatives is implied by rational inattention and why choice probabilities can satisfy an IIA-like condition over consequences while still responding to prior beliefs.
- **Applications.** The model explains context effects such as correlation among payoffs, duplicates, and comovement among alternatives. The bus-versus-train example shows that two similar alternatives may attract attention jointly and can affect the probability of choosing the outside option.
- **Takeaway.** The paper gives a microfoundation for multinomial logit in which random choice is rationally generated by costly attention rather than assumed through additive extreme-value shocks.

2 Incremental contribution of the paper

2.1 Contribution relative to random utility logit

- The classic multinomial logit model is usually derived from additive i.i.d. extreme-value taste shocks or measurement errors.
- Matějka and McKay derive logit-like probabilities from a fully optimizing information-acquisition problem.
- In their model, the same individual with the same utility function can choose probabilistically because she optimally chooses not to learn the state perfectly.
- The stochastic component is therefore not an unexplained preference residual; it is an endogenous consequence of limited attention.

Interpretation: The paper changes the interpretation of logit errors. Logit noise is no longer necessarily taste heterogeneity or econometric error. It can be optimal uncertainty created by information costs.

2.2 Contribution relative to rational inattention

- Earlier rational inattention models often focus on continuous actions or macroeconomic decision problems.
- This paper solves the rational inattention problem in a finite discrete choice environment.
- It shows that entropy costs produce closed-form conditional choice probabilities that are logit in true payoffs.
- It also derives the unconditional probabilities from an auxiliary convex optimization problem.

Interpretation: The paper makes rational inattention operational for discrete-choice economics. It provides a bridge between Sims-style information costs and the empirical logit tradition.

2.3 Contribution relative to discrete-choice demand estimation

- In standard discrete choice, alternative-specific constants are often interpreted as unobserved mean utility or taste shifters.
- In this paper, the corresponding action weights $\alpha_i = \lambda \log P_i^0$ reflect prior beliefs and endogenous attention allocation.
- Products that look promising ex ante attract attention and enter the choice probability with larger weights even before realized payoffs are known.

Interpretation: The paper gives a structural interpretation to alternative-specific constants: they can summarize the probability that the decision maker chooses to pay attention to the alternative.

2.4 Contribution to behavioral and psychological choice theory

- The model formalizes attention allocation without imposing ad hoc consideration sets or search rules.
- It explains why decision makers may ignore dominated or low-prior alternatives.
- It explains how similar alternatives can be treated as substitutes in attention rather than only in utility.
- It rationalizes probabilistic mistakes without assuming irrationality.

Interpretation: The paper provides a disciplined economic foundation for limited attention in choices among discrete alternatives.

3 Model environment

3.1 Actions, states, and payoffs

The decision maker chooses one action from a finite set

$$\mathcal{A} = \{1, \dots, N\}. \quad (2)$$

The state of nature is a payoff vector

$$\mathbf{v} = (v_1, \dots, v_N) \in \mathbb{R}^N, \quad (3)$$

where v_i is the payoff of action i . Before acquiring information, the decision maker has a prior distribution G over payoff vectors.

Interpretation: The state is not a hidden type of the agent. It is the full payoff vector of the choice problem. The decision maker is uncertain about which action is truly best.

3.2 Two-stage decision problem

1. **Information stage.** The decision maker selects an information strategy, represented by a joint distribution between signals and states. This determines posterior beliefs.
2. **Choice stage.** After observing the signal and forming posterior belief B , the decision maker chooses the action with the highest expected payoff:

$$V(B) = \max_{i \in \mathcal{A}} E_B[v_i]. \quad (4)$$

Interpretation: The decision maker is Bayesian after observing information. Imperfect choices arise because information is costly, not because the second-stage action is irrational.

3.3 Information cost

The cost of information is proportional to the expected reduction in Shannon entropy:

$$\lambda [H(G) - E_s H(F(\cdot | s))], \quad (5)$$

where G is the prior, $F(\cdot | s)$ is the posterior induced by signal s , and $\lambda > 0$ is the marginal cost of information.

For a finite state space, entropy is

$$H(P) = - \sum_{\omega} P(\omega) \log P(\omega). \quad (6)$$

Interpretation: Entropy measures how much uncertainty is resolved. Higher λ makes information more expensive and leads the decision maker to rely more heavily on prior beliefs.

3.4 Choice-probability formulation

Rather than solving directly for signals, the paper formulates the problem in terms of conditional choice probabilities $\mathcal{P}_i(\mathbf{v})$.

Let

$$P_i^0 = \int \mathcal{P}_i(\mathbf{v}) dG(\mathbf{v}) \quad (7)$$

be the unconditional probability of choosing action i .

The rational inattention problem can be written as maximizing expected payoff net of information costs:

$$\max_{\{\mathcal{P}_i(\mathbf{v})\}_{i=1}^N} \int \sum_{i=1}^N \mathcal{P}_i(\mathbf{v}) v_i dG(\mathbf{v}) - \lambda \int \sum_{i=1}^N \mathcal{P}_i(\mathbf{v}) \log \left(\frac{\mathcal{P}_i(\mathbf{v})}{P_i^0} \right) dG(\mathbf{v}), \quad (8)$$

subject to

$$\mathcal{P}_i(\mathbf{v}) \geq 0, \quad \sum_{i=1}^N \mathcal{P}_i(\mathbf{v}) = 1. \quad (9)$$

Interpretation: The information cost is a relative entropy term. The decision maker pays when conditional choice probabilities differ from unconditional choice probabilities.

4 Main analytical results

4.1 Theorem 1: conditional choice probabilities

The paper's main theorem states that if $\lambda > 0$, optimal conditional choice probabilities satisfy

$$\mathcal{P}_i(\mathbf{v}) = \frac{P_i^0 e^{v_i/\lambda}}{\sum_{j=1}^N P_j^0 e^{v_j/\lambda}}. \quad (10)$$

If $\lambda = 0$, the decision maker selects payoff-maximizing action(s) with probability one.

Interpretation: Conditional on a state, choice probabilities are logit in realized payoffs. The difference from standard logit is the appearance of P_i^0 , which is itself chosen optimally.

4.2 Relation to the multinomial logit formula

Define

$$\alpha_i = \lambda \log P_i^0. \quad (11)$$

Then

$$\mathcal{P}_i(\mathbf{v}) = \frac{e^{(v_i + \alpha_i)/\lambda}}{\sum_{j=1}^N e^{(v_j + \alpha_j)/\lambda}}. \quad (12)$$

- If $P_i^0 = 1/N$ for all i , then all α_i are equal and disappear from relative probabilities.
- If prior beliefs favor one action, then its P_i^0 is higher and its α_i is higher.
- As λ rises, the decision maker processes less information and P_i^0 becomes more important for choice.

Interpretation: The standard logit model appears as a special case. The generalized model includes endogenous action weights that summarize prior beliefs and attention allocation.

4.3 Alternative formulation: solving for unconditional probabilities

Lemma 2 shows that the conditional probabilities can be obtained from unconditional probabilities $\{P_i^0\}_{i=1}^N$ solving

$$\max_{\{P_i^0\}_{i=1}^N} \int_{\mathbf{v}} \lambda \log \left(\sum_{i=1}^N P_i^0 e^{v_i/\lambda} \right) dG(\mathbf{v}), \quad (13)$$

subject to

$$P_i^0 \geq 0, \quad \sum_i P_i^0 = 1. \quad (14)$$

Interpretation: The high-dimensional information strategy can be reduced to a finite-dimensional optimization over unconditional choice probabilities.

4.4 Normalization condition

For any action i with $P_i^0 > 0$, the solution satisfies

$$\int_{\mathbf{v}} \frac{e^{v_i/\lambda}}{\sum_{j=1}^N P_j^0 e^{v_j/\lambda}} dG(\mathbf{v}) = 1. \quad (15)$$

Interpretation: This condition pins down the unconditional probabilities. It says the unconditional probability assigned to action i must be consistent with the average conditional probability implied by the logit formula.

5 Characterization and IIA

5.1 State-contingent choice probabilities

The paper asks whether the rational-inattention logit formula can be characterized from choice data without knowing the true payoffs or the prior. The key objects are state-contingent choice probabilities over consequences.

Interpretation: This section is important because it turns the model from a purely structural theory into a set of observable restrictions on choice probabilities.

5.2 Standard IIA intuition

The traditional IIA axiom says that for two consequences x and y , their choice-probability ratio is invariant to adding or removing irrelevant alternatives:

$$\frac{\Pr(x \text{ chosen from } A)}{\Pr(y \text{ chosen from } A)} = \frac{\Pr(x \text{ chosen from } B)}{\Pr(y \text{ chosen from } B)}. \quad (16)$$

Interpretation: In ordinary logit, IIA is a restriction on alternatives. In the rational-inattention model, the correct restriction applies to consequences across states after accounting for prior-induced action weights.

5.3 Prior-dependent IIA

Because choice probabilities contain P_i^0 , IIA across actions can fail when prior beliefs change. However, once one conditions correctly on consequences, the model satisfies an IIA-like representation.

Interpretation: This is a subtle but important point. The rational inattention model generates logit probabilities conditional on state, but prior beliefs can shift unconditional probabilities and therefore alter apparent IIA patterns over actions.

5.4 Dominated and duplicate actions

The paper shows that if one action is dominated by another in all states, it receives zero unconditional probability and is never selected. It also studies duplicate or highly correlated actions.

Interpretation: Rational inattention naturally eliminates alternatives that are never worth attention. It also predicts that similar alternatives may share attention, leading to substitution patterns different from naive logit.

6 Examples and economic intuition

6.1 Bus versus train payoff example

Table 1 considers a decision maker choosing between a red bus, a blue bus, and a train. The state determines the payoffs of each vehicle. The parameter ρ controls correlation between the payoffs of the two buses, and R is the train payoff.

Interpretation: The example shows that the decision maker may choose to learn about one dimension of uncertainty rather than all payoffs. If the two buses have highly correlated payoffs, the decision maker can treat them like one effective alternative.

6.2 State-contingent consequences example

Table 2 gives a consequence-level example with three actions and three states. The consequences are goods such as a mug, pen, T-shirt, or nothing.

Interpretation: This example motivates the characterization of choice probabilities through consequences rather than actions. It clarifies how IIA should be interpreted when actions produce different consequences in different states.

6.3 Bus-choice probability figure

Figure 1 shows the unconditional probability of selecting a bus as a function of the correlation parameter ρ for different information costs λ .

- When the buses' payoffs are perfectly correlated, they become duplicates and the probability of selecting a bus approaches one-half.
- When the two bus payoffs are less correlated, the decision maker may devote attention to the bus options because one of them is more likely to be best.
- Higher information cost changes the extent to which the decision maker differentiates between the two buses.

Interpretation: The figure illustrates how attention substitutes among similar alternatives. Similarity affects choice not only through payoffs but through the value of learning about the alternatives.

7 Identification and empirical interpretation

7.1 What can be estimated

In an empirical discrete-choice model, the rational-inattention foundation suggests estimating

$$\Pr(i \mid \mathbf{v}) = \frac{e^{(v_i + \alpha_i)/\lambda}}{\sum_j e^{(v_j + \alpha_j)/\lambda}}, \quad (17)$$

where α_i reflects prior beliefs and endogenous attention.

Interpretation: Standard logit estimation can be interpreted as estimating payoff sensitivity and attention-weighted priors, not merely preferences and exogenous shocks.

7.2 What changes relative to ordinary MNL

- The logit scale λ is an information-cost parameter.
- Alternative-specific constants can be attention weights.
- Changes in prior information can alter choice probabilities even when true payoffs do not change.
- Similarity and correlation among alternatives affect the value of information and thus the unconditional probabilities.

Interpretation: The same mathematical logit formula can support a very different economic interpretation.

7.3 Limits of the approach

- The entropy cost function is tractable but may not capture every cognitive constraint.
- The model abstracts from sequential search costs, memory constraints, and attention dynamics across repeated choices.
- The prior distribution G is critical but often hard to observe directly.
- With multiple solutions for unconditional probabilities, special conditions are needed for uniqueness.

Interpretation: The paper provides a foundation, not a complete empirical recipe for all attention problems.

8 Tables and figures

8.1 Table 1 and Table 2: motivating examples

Read:

- Table 1 gives the payoff matrix for the red-bus, blue-bus, and train example.
- The parameter ρ controls how the bus payoffs comove across states.
- Table 2 gives a state-contingent consequence matrix used for the characterization and IIA discussion.
- The examples show why it matters whether one focuses on actions, states, or consequences.

Economic message: The two tables motivate the paper’s central insight: attention is allocated to payoff-relevant uncertainty, and the resulting choice probabilities depend on prior structure and payoff comovement.

TABLE 1—PAYOFF MATRIX (Each column lists v for that state)

	State 1	State 2	State 3	State 4
Action 1: red bus	0	1	0	1
Action 2: blue bus	0	0	1	1
Action 3: train	R	R	R	R
$G(v)$	$\frac{1}{4}(1 + \rho)$	$\frac{1}{4}(1 - \rho)$	$\frac{1}{4}(1 - \rho)$	$\frac{1}{4}(1 + \rho)$

Table 1: payoff matrix

TABLE 2—STATE-CONTINGENT CONSEQUENCES

	State 1	State 2	State 3
Action 1	Mug	Pen	Nothing
Action 2	Mug	Pen	Pen
Action 3	Nothing	T-shirt	T-shirt

Table 2: state-contingent consequences

8.2 Theorem 1 and Lemma 2: solution of the rational-inattention problem

Read:

- Theorem 1 gives the conditional logit formula with endogenous unconditional probabilities P_i^0 .
- Lemma 2 reduces the full information-choice problem to an optimization over P_i^0 .
- These results jointly show that the decision maker’s entire optimal attention strategy can be summarized by action weights.

Economic message: This is the mathematical core of the paper: rational attention plus entropy costs yields logit choice probabilities.

A. Solving for Choice Probabilities

We begin our analysis of the model with a general analytical expression for the probability that the decision maker chooses a particular action conditional on the state of nature.

THEOREM 1: *If $\lambda > 0$, then the optimal strategy has conditional choice probabilities that satisfy*

$$(13) \quad \mathcal{P}_i(v) = \frac{\mathcal{P}_i^0 e^{v_i/\lambda}}{\sum \mathcal{P}_i^0 e^{v_i/\lambda}} \quad \text{almost surely.}$$

Theorem 1: conditional logit formula

fixed, the problem is equivalent to the one with the original payoffs and the information cost scaled by $1/c$. By scaling up the payoffs, one is scaling up the differences between them and therefore raising the stakes for the decision maker. The decision maker chooses to process more information because more is at stake and thus is more likely to select the action that provides the highest payoff. The decision maker behaves just as he would if the cost of information had fallen.

Equation (13) does not give a fully explicit expression for the choice probabilities because it depends on the $\mathcal{P}_i^0$ terms, which are themselves part of the decision maker’s strategy although they are independent of the state. We can substitute equation (13) into the objective function to arrive at the following formulation of the optimization problem that we will use to solve for the unconditional choice probabilities.

LEMMA 2 (Alternative Formulation): *The collection of conditional probabilities $\mathcal{P} = \{\mathcal{P}_i(v)\}_{i=1}^N$ solves (10)–(12) in Corollary 1 if and only if the probabilities are given by (13) with the unconditional probabilities, $\{\mathcal{P}_i^0\}_{i=1}^N$, that solve*

$$(14) \quad \max_{\{\mathcal{P}_i^0\}_{i=1}^N} \int v \cdot \lambda \log \left(\sum_{i=1}^N \mathcal{P}_i^0 e^{v_i/\lambda} \right) G(dv),$$

Lemma 2: alternative formulation

8.3 Corollary 2 and Figure 1: normalization and the bus example

Read:

- Corollary 2 states the normalization condition that determines unconditional probabilities.
- Figure 1 plots the unconditional probability of selecting a bus for different ρ and λ values.

- The probability of choosing a bus falls when the two buses become closer duplicates.
- Information costs change how much the decision maker distinguishes among similar alternatives.

Economic message: The figure turns the theory into an intuitive comparative static: correlated alternatives reduce the value of learning separately about them.

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THIS NOVEL FORMULATION IS USEFUL FOR TWO REASONS. FIRST, IT ALLOWS FOR GREATER INSIGHTS INTO THE PROPERTIES OF CHOICE PROBABILITIES THAN THE ORIGINAL PROBLEM AND WE USE IT EXTENSIVELY IN THE PROOFS IN SECTION IV. SECOND, IT REDUCES GREATLY THE COMPLEXITY OF THE OPTIMIZATION PROBLEM AND ALLOWS FOR MORE EFFICIENT COMPUTATIONS. RATIONAL INATTENTION PROBLEMS WITH CONTINUOUS CHOICE VARIABLES CAN ALSO BE FORMULATED THIS WAY.¹⁴ THE FIRST-ORDER CONDITIONS OF THIS PROBLEM GIVE US

COROLLARY 2 (Normalization Condition): For all i such that $P_i^0 > 0$, the solution satisfies

$$P_i^0 = \frac{e^{\alpha_i / \lambda}}{\sum_j e^{\alpha_j / \lambda}}$$

Corollary 2: normalization condition

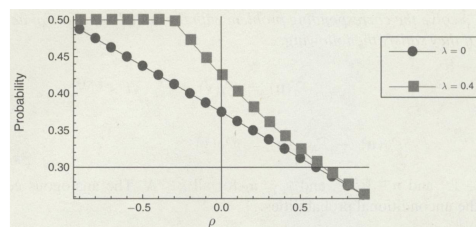


FIGURE 1. UNCONDITIONAL PROBABILITY OF SELECTING A BUS FOR VARIOUS VALUES OF λ AND ρ (The probability is the same for both the red and blue buses)

Figure 1: unconditional bus-choice probability

9 Short summary

- The paper microfound multinomial logit using rational inattention.
- The decision maker chooses both what information to process and which action to take.
- Information is costly, with cost proportional to entropy reduction.
- Optimal conditional choice probabilities are logit in realized payoffs.
- The logit weights include endogenous unconditional choice probabilities P_i^0 .
- When unconditional probabilities are uniform, the model reduces to standard multinomial logit.
- The logit scale parameter λ is the marginal cost of information.
- Prior beliefs affect choice through the action weights $\alpha_i = \lambda \log P_i^0$.
- The model gives a new interpretation of IIA and shows how similarity among alternatives affects attention.
- The framework explains stochastic choice as rationally chosen ignorance rather than exogenous taste noise.

10 Conclusion

10.1 Core conclusion

Matějka and McKay show that the multinomial logit formula can arise from optimal information acquisition in a discrete-choice problem. A rationally inattentive decision maker chooses a limited information strategy, then chooses the action with the highest posterior expected payoff. The resulting conditional choice probabilities have a logit form.

10.2 Economic intuition

Information is valuable because it helps the decision maker identify the best action. But information is costly, so the decision maker does not learn everything. More attractive alternatives receive more attention and higher unconditional choice probability. The model therefore produces probabilistic choice and logit-like behavior even without preference shocks.

10.3 Takeaway

The paper gives the multinomial logit model a new foundation. It transforms logit from a statistical error model into a theory of optimal limited attention. This is important for interpreting logit estimates, alternative-specific constants, scale parameters, and substitution patterns in discrete-choice applications.